SPRING FRAMEWORK

CONFIGURATION SPRING

- XML CONFIGURATION
- JAVA BASED CONFIGURATION (avec @Configuration et @Bean)
- ANNOTATION CONFIGURATION (component scanning)

XML CONFIGURATION

On définit les beans et leurs dépendances dans un fichier XML .

Fichiers jar utilisés :

- commons-logging-X.X.jar
- spring-beans-X.X.X.jar
- spring-context-X.X.X.jar
- spring-core-X.X.X.jar
- spring-expression-X.X.X.jar

Student.java

```java
public class Student {

    private String name;
    private int number;
    private String email;

    //Getters and setters

    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int getNumber() {
        return number;
    }
    public void setNumber(int number) {
        this.number = number;
    }
    public String getEmail() {
        return email;
    }
    public void setEmail(String email) {
        this.email = email;
    }

    public void afficher() {
        System.out.println("Nom : " + name + " - Numero : " + number + " - Email : " + email);
    }
}
```

Main.java

```java
public class MainApp {

    public static void main(String[] args) {

        ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
        Student std = (Student) context.getBean("student");
        std.afficher();

    }
}
```

applicationContext.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">

        <bean id="student" class="com.tuto.Student">
            <property name="name" value="John"/>
            <property name="number" value="1"/>
            <property name="email" value="john@gmail.com"/>
        </bean>

</beans>
```

JAVA BASED CONFIGURATION

Fichiers jar utilisés :

- commons-logging-X.X.jar
- spring-aop.X.X.X.jar
- spring-beans-X.X.X.jar
- spring-context-X.X.X.jar
- spring-core-X.X.X.jar
- spring-expression-X.X.X.jar

Personne.java

```java
package com.tuto;

import org.springframework.stereotype.Component;

@Component
public class Personne {

    private String firstName;
    private String lastName;
    private int age;

    public String getFirstName() {
        return firstName;
    }
    public void setFirstName(String firstName) {
        this.firstName = firstName;
    }
    public String getLastName() {
        return lastName;
    }
    public void setLastName(String lastName) {
        this.lastName = lastName;
    }
    public int getAge() {
        return age;
    }
    public void setAge(int age) {
        this.age = age;
    }


    public void affiche() {
        System.out.println("Nom : " +firstName);
        System.out.println("Prenom : " +lastName);
        System.out.println("Age : " +age);
    }
}
```

AppConfig.java

```java
package com.tuto;

import org.springframework.context.annotation.Bean;

@Configuration
public class AppConfig {

    @Bean
    public Personne personne() {

        Personne pers = new Personne();

        pers.setFirstName("John");
        pers.setLastName("Doe");
        pers.setAge(30);

        return pers;
        }

}
```

Main.java

```java
package com.tuto;

import org.springframework.context.ApplicationContext;

public class MainApp {

    public static void main(String[] args) {

        //Spring Container is created
        ApplicationContext context =
                new AnnotationConfigApplicationContext(AppConfig.class);

        Personne p = context.getBean(Personne.class);
        p.affiche();
    }
}
```

ANNOTATION CONFIGURATION

Fichiers jar utilisés :

- commons-logging-X.X.jar
- spring-aop.X.X.X.jar
- spring-beans-X.X.X.jar
- spring-context-X.X.X.jar
- spring-core-X.X.X.jar
- spring-expression-X.X.X.jar

Student.java

```java
@Component
public class Student {

    @Value("John")
    private String name;

    @Value("10")
    private int number;

    @Value("john@mail.com")
    private String email;


    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int getNumber() {
        return number;
    }
    public void setNumber(int number) {
        this.number = number;
    }
    public String getEmail() {
        return email;
    }
    public void setEmail(String email) {
        this.email = email;
    }

    public void display() {
        System.out.println("Name : " +name);
        System.out.println("No : " +number);
        System.out.println("Email : " +email);
    }
}
```

AppConfig.java

```java
package com.tuto;

import org.springframework.context.annotation.ComponentScan;

@Configuration
@ComponentScan(basePackages = "com.tuto")
public class AppConfig {

}
```

Main.java

```java
package com.tuto;

import org.springframework.context.ApplicationContext;

public class MainApp {

    public static void main(String[] args) {

        ApplicationContext context =
                new AnnotationConfigApplicationContext(AppConfig.class);

        Student std = context.getBean(Student.class);
        std.display();

    }

}
```